Rethinking Spatial Units for BEV Fast-Charging Infrastructure

Arterial Mobility Zoning

- Extract Major Arterial Roads
- Generate Arterial Zones

Ridge-Path Extraction

- Build Graph
- Compute Weights
- Extract Ridge-Paths

Null-Model Benchmarking

- Generate Random Layouts
- Compute Alignment

Planning Insight Out

- Identify BEV Activity Backbone
- Evaluate Charging Network
- Provide Recommendations

BEV Activity Mapping

- BEV Ownership Administrative Zones
- Re-map to Arterial Zones

Charger Alignment Evaluation

- Compute WPD
- Measure Accessibility

Methodology

Results

Metric	Value
Empirical Evaluation	
FLO DCFC network proximity metric ($\mathrm{WPD}_{\mathrm{FLO}}$)	2.8820
Total random experiments (R)	1050
Random experiments with lower WPD	902
Empirical percentile rank (%)	85.74
Mean of random metrics ($\mathrm{E}[\mathrm{WPD}_{\mathrm{null}}]$)	2.1272
Standard deviation (σ_{null})	0.6767
Minimum / Maximum random WPD_{null}	0.7471 / 4.707

Conclusion: Current FLO fast charging network performs no better than random placement, revealing a lack of meaningful mesoscale spatial optimization in its deployment.